

Winlogbeat & ELK Stack Lab

Windows Event Collection & SOC Dashboard with Winlogbeat, Elasticsearch, and Kibana

Prepared by

Chontele Coleman

Introduction

The purpose of this lab is to simulate a Security Operations Center (SOC) environment. Tools I implemented are Kali Linux, Windows 10 VM, Windows Server Domain Controller VM, Winlogbeat, Elasticsearch, and Kibana.

Scope

This project focuses on building a centralized log collection and visualization environment for Windows Security Events. Using **Winlogbeat** on multiple Windows endpoints (Windows 10 workstation and Windows Server Domain Controller), logs are forwarded into an **Elasticsearch** cluster and visualized in **Kibana** dashboards.

Lab Environment Setup 1.1

Kali Linux VM:

- Installed Elasticsearch 8.19.2
- Installed Kibana
- Verified Elasticsearch & Kibana are running:
 - `sudo systemctl status elasticsearch`
 - `sudo systemctl status Kibana`
- Verified Kibana web interface was accessible:
 - <http://192.168.56.110:5601>.
- Generated an enrollment token and verification code:

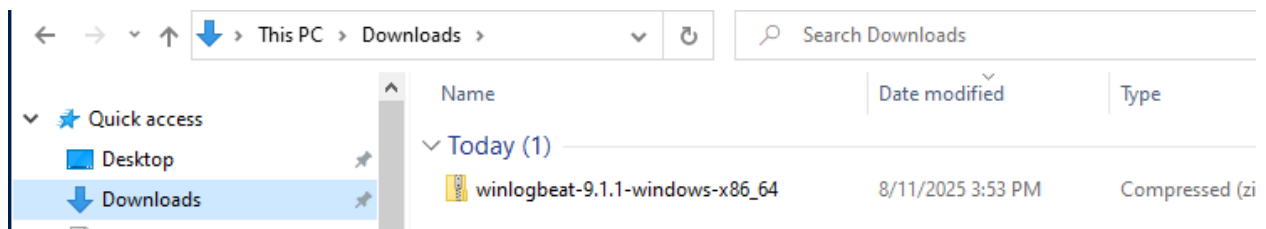
```
(chonteale@Kali)-[~]  
$ sudo systemctl start elasticsearch  
[sudo] password for chonteale:  
  
(chonteale@Kali)-[~]  
$ systemctl status elasticsearch  
● elasticsearch.service - Elasticsearch  
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: disabled)  
   Active: active (running) since Wed 2025-09-03 18:28:41 EDT; 2min 41s ago
```

```
File  Actions  Edit  View  Help  
  
(chonteale@Kali)-[~]  
$ sudo systemctl start kibana  
  
(chonteale@Kali)-[~]  
$ systemctl status kibana  
● kibana.service - Kibana  
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: disabled)  
   Active: active (running) since Wed 2025-09-03 18:27:21 EDT; 5min ago  
   Invocation: 005fbc28c6b0/525b50/0dc00001/07c
```


Lab Environment Setup 1.2

Windows 10 & Domain Controller VMs:

- Installed Winlogbeat 9.1.1 on both VMs
- Configured to forward logs to Elasticsearch on Kali VM
- Services monitored:
 - Application
 - Security
 - System
 - Setup
 - Forwarded Events
- Configured winlogbeat.yml
- Confirmed service running in Windows
 - Start-Service winlogbeat
 - Get-Service winlogbeat



```
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> cd "C:\Program Files\winlogbeat-9.1.1-windows-x86_64"
PS C:\Program Files\winlogbeat-9.1.1-windows-x86_64> dir

Directory: C:\Program Files\winlogbeat-9.1.1-windows-x86_64

Mode                LastWriteTime         Length Name
----                -
d-----         8/11/2025   4:08 PM            kibana
d-----         8/11/2025   4:08 PM            module
-a-----         8/11/2025   4:08 PM             41 .build_hash.txt
-a-----         8/11/2025   4:08 PM          413739 fields.yml
-a-----         8/11/2025   4:08 PM          2713 install-service-winlogbeat.ps1
-a-----         8/11/2025   4:08 PM          13675 LICENSE.txt
-a-----         8/11/2025   4:08 PM         4409315 NOTICE.txt
-a-----         8/11/2025   4:08 PM           835 README.md
-a-----         8/11/2025   4:08 PM          298 uninstall-service-winlogbeat.ps1
-a-----         8/11/2025   4:08 PM        112953576 winlogbeat.exe
-a-----         8/11/2025   4:08 PM          67512 winlogbeat.reference.yml
-a-----         8/11/2025   4:08 PM          7553 winlogbeat.yml
```

```
PS C:\Program Files\winlogbeat-9.1.1-windows-x86_64> notepad .\winlogbeat.yml

winlogbeat - Notepad
File Edit Format View Help
winlogbeat.event_logs:
  - name: Security
    ignore_older: 72h
    event_id: 4624,4625,4634,4648,4672,4720,4722,4723,4724,4725,4726

# Output to a dedicated folder to avoid confusion
output.file:
  path: "C:/winlogbeat-logs"
  filename: "security_logs"
  rotate_every_kb: 10000
  number_of_files: 5
  permissions: 0600

# Disable Elastic and Logstash outputs
output.elasticsearch:
  enabled: false
output.logstash:
  enabled: false

# Logging for debugging (goes to same folder)
logging.level: info
logging.to_files: true
```

```
Select Administrator: Windows PowerShell
C:\Program Files\winlogbeat-9.1.1-windows-x86_64> .\winlogbeat.exe test config -c .\winlogbeat.yml -e
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.517-0400","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":826},"message":"Home path: [C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64] Config path: [C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64] Data path: [C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\data] Logs path: [C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\logs]","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.517-0400","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":834},"message":"Beat ID: 4c5cf219-c94e-4c47-8360-14a1db90e3c","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.553-0400","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).createBeater","file.name":"instance/beat.go","file.line":1331},"message":"Setup Beat: winlogbeat; Version: 9.1.1 (FIPS distribution: false)","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.555-0400","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1384},"message":"Beat info","service.name":"winlogbeat","system_info":{"beat":{"path":{"config":"C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64","data":"C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\data"},"home":"C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64","logs":"C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\logs"},"type":"winlogbeat"},"uid":"4c5cf219-c94e-4c47-8360-14a1db90e3c"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.556-0400","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1393},"message":"Build info","service.name":"winlogbeat","system_info":{"build":{"commit":"1292cd58f48325c041317d9a8bdc1f1875bc6cf5f","libbeat":"9.1.1","time":"2025-08-04T23:58:41.000Z"},"version":"9.1.1"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.556-0400","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1396},"message":"Go runtime info","service.name":"winlogbeat","system_info":{"go":{"os":"windows","arch":"amd64","max_procs":2,"version":"go1.24.4"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.563-0400","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1402},"message":"Host info","service.name":"winlogbeat","system_info":{"host":{"architecture":"x86_64","native_architecture":"x86_64","boot_time":"2025-08-11T14:43:28-04:00","name":"Windows10-Admin","ip":"fd17:625c:f037:2:84c9:4fca:b945:b723","fd17:625c:f037:2:7c1e:1cba:cdcf:ed4b","fe80::84c9:4fca:b945:b723","10.0.2.15","fe80::4d39:70b4:1ff9:2bc","192.168.56.108"},"":{"1","127.0.0.1"},"kernel_version":"10.0.19041.2006 (WinBuild.160101.0800)","mac":["08:00:27:95:5a:bf","08:00:27:4a:54:80"],"os":{"type":"windows","family":"windows","platform":"windows","name":"Windows 10 Enterprise Evaluation","version":"10.0","major":10,"minor":0,"patch":0,"build":"19045.2006"},"timezone":"EDT","timezone_offset_sec":-14400,"id":"0af4b5af-d299-4701-a223-dfd34dfalbb5"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.564-0400","log.logger":"beat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1431},"message":"Process info","service.name":"winlogbeat","system_info":{"process":{"cwd":"C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64","exe":"C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\winlogbeat.exe","name":"winlogbeat.exe","pid":740,"ppid":6456,"start_time":"2025-08-11T16:17:30.744-0400"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.611-0400","log.logger":"file","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1402},"message":"Initialized file output. path=C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\logs\\security_logs max_size_bytes=10240000 max_backups=5 permissions=-rw-rw-r--","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.623-0400","log.logger":"publisher","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/publisher/pipeline.LoadWithSettings","file.name":"pipeline/module.go","file.line":105},"message":"Beat name: Windows10-Admin","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2025-08-11T16:17:31.623-0400","log.logger":"winlogbeat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/publisher/pipeline.LoadWithSettings","file.name":"pipeline/module.go","file.line":105},"message":"State will be read from and persisted to C:\\Program Files\\winlogbeat-9.1.1-windows-x86_64\\data\\winlogbeat.yml","service.name":"winlogbeat","ecs.version":"1.6.0"}
onfig OK
C:\Program Files\winlogbeat-9.1.1-windows-x86_64>
```

```
Microsoft Edge PS C:\Program Files\winlogbeat-9.1.1-windows-x86_64> .\install-service-winlogbeat.ps1
[SC] DeleteService SUCCESS

Status Name DisplayName
-----
Stopped winlogbeat winlogbeat

PS C:\Program Files\winlogbeat-9.1.1-windows-x86_64> Start-Service winlogbeat
PS C:\Program Files\winlogbeat-9.1.1-windows-x86_64> Get-Service winlogbeat

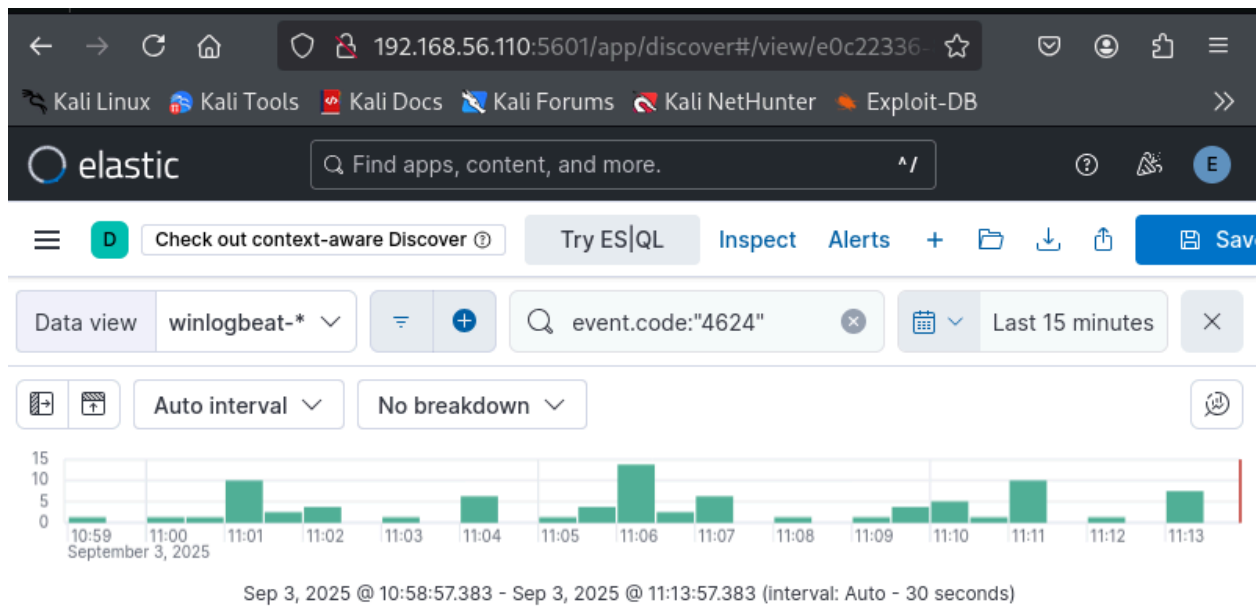
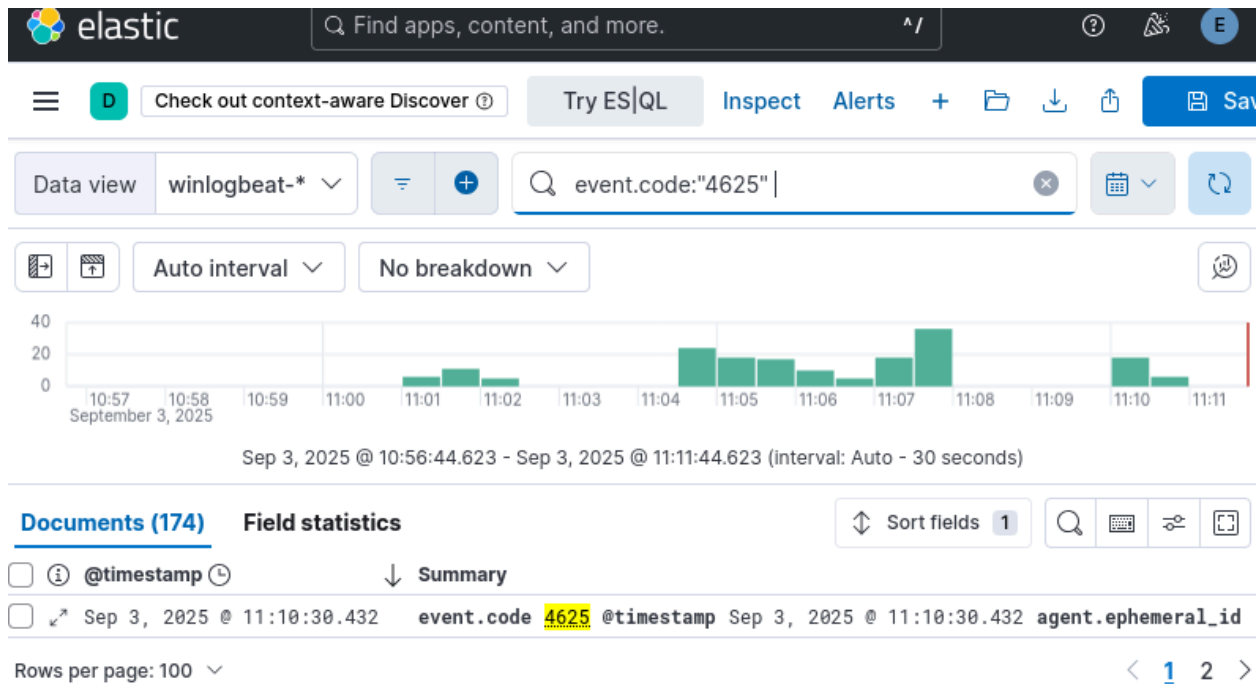
Status Name DisplayName
-----
Running winlogbeat winlogbeat
```

Event Collection, Validation, & Verification

Security Events Captured:

- 4625 – Failed logon attempts
- 4624 – Successful logons

- 4688 – Process creation
- 7045 – Service creation (captured but limited details due to SYSTEM context).



Loading documents

✓ Discover session 'Successful Logons' was saved

Check out context-aware Discover ⓘ Try ES|QL Inspect Alerts + Save

Data view winlogbeat-*

event.code:"4688"



Auto interval

No breakdown



Sep 3, 2025 @ 10:59:57.090 - Sep 3, 2025 @ 11:14:57.090 (interval: Auto - 30 seconds)

Documents (93) Field statistics

Sort fields 1

@timestamp

Summary

Sep 3, 2025 @ 11:13:36.542 event.code 4688 @timestamp Sep 3, 2025 @ 11:13:36.542 agent.ephemeral_id

Rows per page: 100

Check out context-aware Discover ⓘ Try ES|QL Inspect Alerts + Save

Data view winlogbeat-*

event.code:"7045"



Auto interval

No breakdown



Sep 3, 2025 @ 11:01:57.236 - Sep 3, 2025 @ 11:16:57.236 (interval: Auto - 30 seconds)

Find in table

Documents (6) Field statistics

Sort fields 1

@timestamp

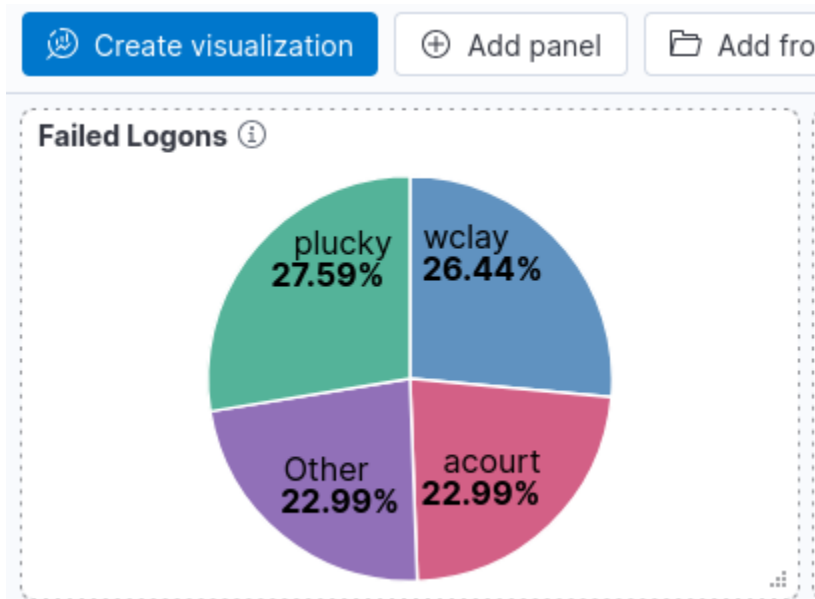
Summary

Sep 3, 2025 @ 11:06:57.216 event.code 7045 @timestamp Sep 3, 2025 @ 11:06:57.216 agent.ephemeral_id 8399134c-533a-40e7-91a4-f0c38c18af94 agent.hostname WIN10-WS19 agent.id

Visualization Setup

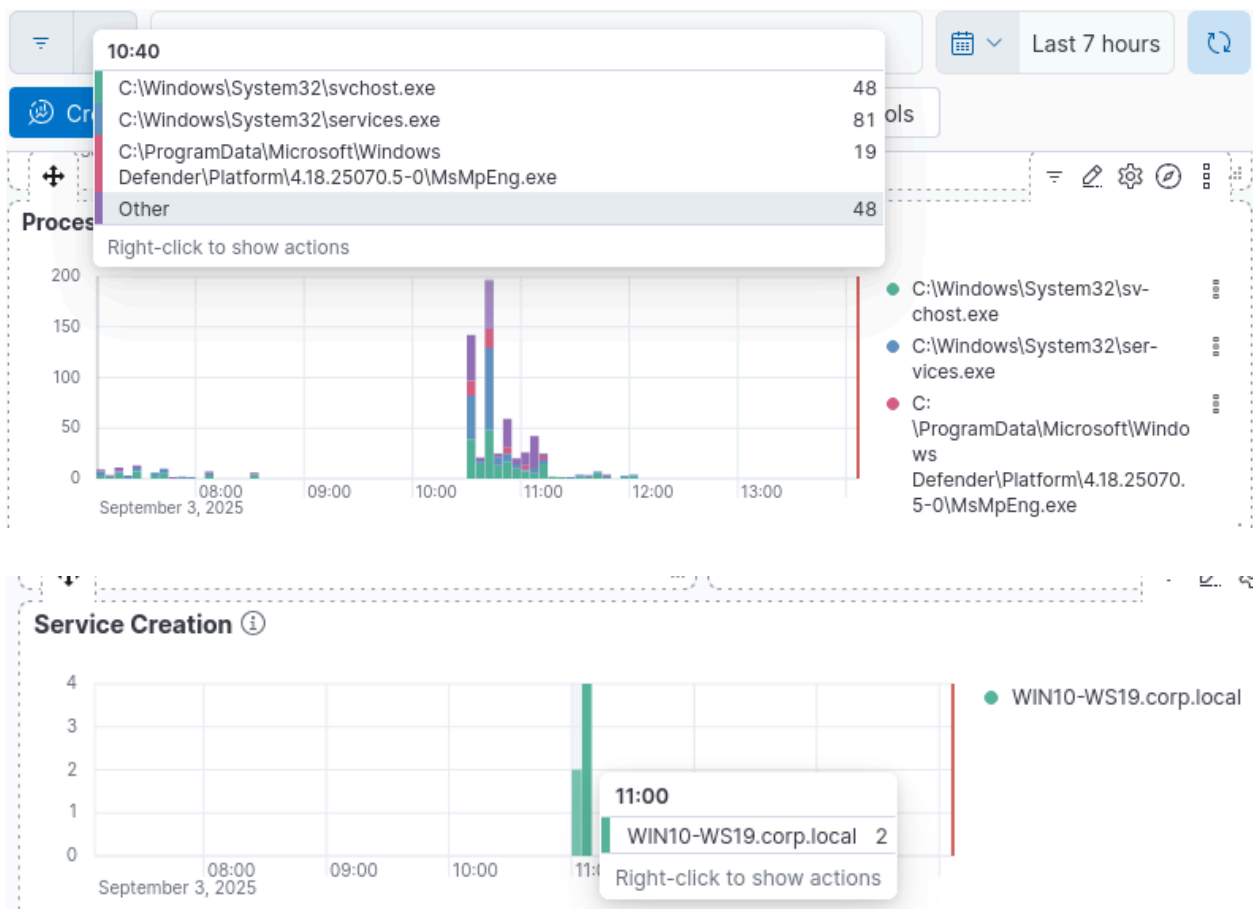
Converted saved searches into visualizations:

- 4625: Pie chart of failed logons over time.
- 4624: Table chart of successful logons per host.
- 4688: Vertical bar chart showing process creation activity.
- 7045: Bar chart of service creation activity by host (limited details due to SYSTEM context).



Successful logons ⓘ

Top 3 values of winlog.event_data.Target	Count of records
DC01\$	384
SYSTEM	106
WIN10-WS19\$	26
Other	47



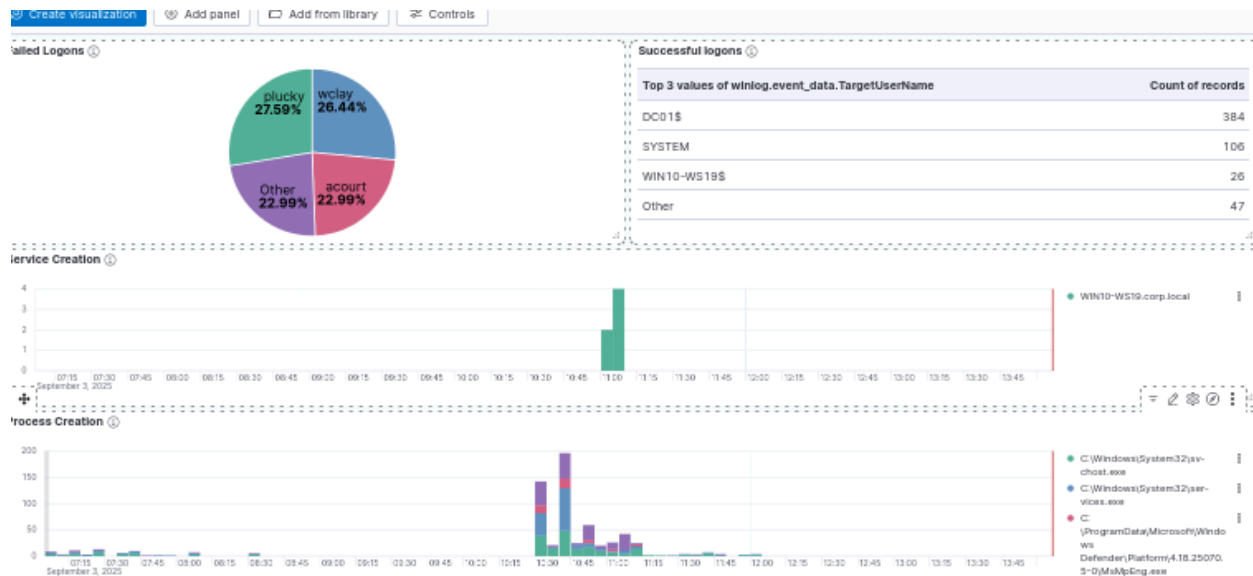
Dashboard Design

- Top Row – Authentication Activity
 - Failed Logins (4625)
 - Successful Logins (4624)
- Middle Row – Execution & Persistence
 - Process Creation (4688)
- Bottom Row – Persistence
 - Service Creation (7045)

Example Workflow

- Analysts can filter by event.code, host.name, or user.name.
- Failed logins (4625) highlight potential brute force attempts.
- Process Creation (4688) can help detect malicious or unusual execution methods.
- Service Creation (7045) shows potential persistence mechanisms.

Dashboard layout



Challenges & Observations

- **Username:** Many events shown as SYSTEM, ADMINISTRATOR, or machine accounts (DC01\$) due to script execution context.
- **7045 Service Creation:** process.name and service.name fields did not populate, limiting visibility.
- **Workaround:** Visualizations were still useful for event trends and SOC-style analysis.

Conclusion

- Successfully built a functional **SOC monitoring dashboard** using Winlogbeat, Elasticsearch, and Kibana.
- Demonstrated collection and visualization of **Windows Security Events**.
- Verified end-to-end data flow from Windows VMs → Winlogbeat → Elasticsearch → Kibana.

Future Improvements

- Enable **Sysmon** for richer logging (process hashes, network connections).
- Configure **Advanced Audit Policies** in Windows for more detailed events.
- Add **alerting** in Kibana (or ElastAlert) for detection of brute force, suspicious processes, or service creation.
- Expand lab with **phishing or malware simulation** for more realistic SOC use cases